

A. LCM Challenge

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

Some days ago, I learned the concept of LCM (least common multiple). I've played with it for several times and I want to make a big number with it.

But I also don't want to use many numbers, so I'll choose three positive integers (they don't have to be distinct) which are not greater than n . Can you help me to find the maximum possible least common multiple of these three integers?

Input

The first line contains an integer n ($1 \leq n \leq 10^6$) — the n mentioned in the statement.

Output

Print a single integer — the maximum possible LCM of three not necessarily distinct positive integers that are not greater than n .

Examples

input	Copy
9	
output	Copy
504	

input	Copy
7	
output	Copy
210	

Note

The least common multiple of some positive integers is the least positive integer which is multiple for each of them.

The result may become very large, 32-bit integer won't be enough. So using 64-bit integers is recommended.

For the last example, we can chose numbers 7, 6, 5 and the LCM of them is $7 \cdot 6 \cdot 5 = 210$. It is the maximum value we can get.

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Round 146 (Div. 1)

Finished

→ Virtual participation

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→ Problem tags

number theory *1600

No tag edit access

→ Contest materials

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